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Research paper

Fecal adrenal hormone patterns during ovulatory and non-ovulatory reproductive cycles in female veiled chameleons (*Chamaeleo calyptratus)*

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ABSTRACT

The relationship between the reproductive (hypothalamic-pituitary-gonadal; HPG) and adrenal (hypothalamic-pituitary-adrenal; HPA) hormone axes is complex and can vary depending on the species and environmental factors affecting an individual. In an effort to understand this relationship in female veiled chameleons (*Chamaeleo calyptratus*), the patterns of fecal metabolites of corticosterone (C), estradiol (E), testosterone (T), and progesterone (P) were analyzed by enzyme immunoassay (EIA) during ovulatory (OC; eggs laid) and non-ovulatory cycles (NOC; no eggs laid). Glucocorticoid (GC) metabolites in the fecal extracts were characterized by HPLC and corticosterone EIA performance was assessed by parallelism, accuracy and precision tests. The results indicated that the assay chosen reliably measured the hormone metabolites present in the fecal extracts. Regular, cyclical hormone metabolite patterns consisting of an E peak followed by peaks of T, P and C in close succession were observed during both ovulatory and non-ovulatory cycles; relative levels of P and C, however, were higher during ovulatory cycles. Corticosterone metabolite levels, in particular, increased throughout vitellogenesis and peaked in late vitellogenesis (in non-ovulatory cycles) or around the time of ovulation, and remained elevated throughout the gravid period, falling just prior to oviposition. The results provide evidence of variation in glucocorticoid production throughout different stages of the reproductive cycle, including a role in the ovulatory process; the physiology, however, remains unclear.

1. Introduction

The relationship between glucocorticoids and reproduction is complex, and highly context-dependent (Moore and Jessop, 2003). While some studies failed to show an increase in plasma corticosterone during the breeding season, such as in bearded dragon lizards; *Pogona barbata* (Amey and Whittier, 2000), and New Zealand common geckos; *Hoplodactylus maculatus* (Girling and Cree, 1995), many reptiles displayed concurrent elevations in plasma corticosterone throughout the breeding season compared to the non-breeding season as observed in side-blotched lizards; *Uta stansburiana* (Wilson and Wingfield, 1992), red-sided garter snakes; *Thamnophis sirtalis parietalis* (Moore et al., 2001), eastern fence lizards; *Sceloporus undulatus* (John-Alder et al., 2002), and Cuban crocodiles; *Crocodylus rhombifer* (Augustine et al., 2020). More specifically, glucocorticoid levels have been shown to vary with the different stages of the reproductive cycle (i.e. previtellogenesis, vitellogenesis, and gravidity), as seen in *U. stansburiana* (Wilson and Wingfield, 1992), desert grassland whiptail lizard; *Cnemidophorus uniparens* (Grassman and Crews, 1990), tuatara; *Sphenodon punctatus* (Tyrrell and Cree, 1998), and the viviparous lizard; *Lacerta vivipara* (Dauphin-Villemant et al., 1990). Therefore, increased glucocorticoid levels can either mediate or be a consequence of normal reproduction (behaviorally and physiologically) (Moore and Jessop, 2003).

Woodley and Moore (2002) found that corticosterone concentration was positively correlated with ovarian weight in vitellogenic tree lizards (*Urosaurus ornatus*) and proposed that it may have a facilitative role in the reproductive cycle of this species. Wilson and Wingfield (1992) demonstrated similar results in side-blotched lizards (*U. stansburiana*), with corticosterone levels increasing with vitellogenesis and remaining elevated during the gravid period. Likewise, Gobbetti et al. (1995)

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